

FACULTY OF ENGINEERING AND BASIC SCIENCES
ACADEMIC PROGRAM: DATA ENGINEERING AND ARTIFICIAL INTELLIGENCE

COURSE: ETL (G01)
Workshop-1: Data Engineer

Introduction

This workshop simulates a real job interview code challenge so it will give you a better understanding about a recruitment process and can be upload to github as a project to start creating a portfolio for your future work life.

Your task is to design and implement an end-to-end ETL process: extract data from a CSV file, transform it into a dimensional data model (DDM), load it into a Data Warehouse (DW), and finally build reports with KPIs and visualizations that query the DW (not the CSV).

A few things to consider:

1. I ask that you complete this challenge within the timeframe agreed on our conversation.
2. I highly recommend not using tools such as Copilot, chatGPT, Claude, or similar to simplify or generate code to support the challenge. Doing this will be grounds for automatic disqualification.

Getting started

In this challenge, you will receive a CSV file with 50,000 rows of candidates data who participated in selection processes (these data were randomly generated), and you will have to do some analysis and manipulations on top of this data.

What is expected

I expect that you get the CSV file and create an application to migrate the data to a relational database. Also, you will display those data from the database in chart visualizations; remember, the data should be stored in a database, and your reports must come from the database, not the CSV file.

In summary, your goal is to:

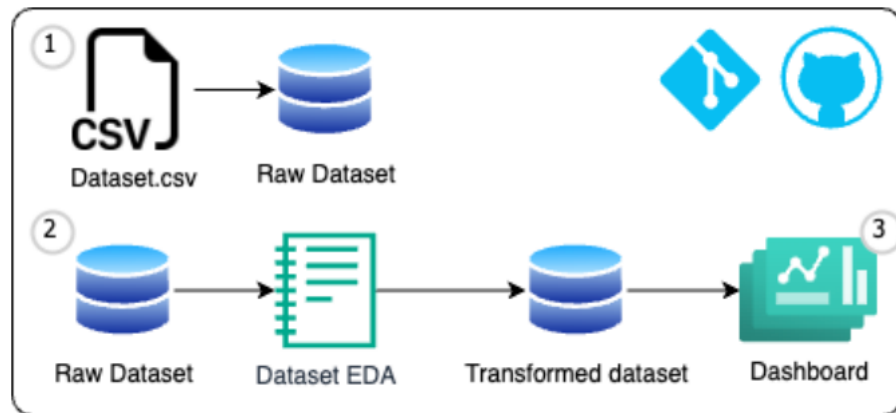
1. Design a Dimensional Data Model (Star Schema).
2. Load the transformed data into a Data Warehouse (DW).
3. Build queries, KPIs, and visualizations directly from the DW.

Technologies

We expect you to use in this challenge:

- Python
- Jupyter Notebook
- Database (you choose)

Diagram



Data description

You will receive a CSV file containing 50k rows with the following fields:

- First Name
- Last Name
- Email
- Country
- Application Date
- YOE (years of experience)
- Seniority
- Technology
- Code Challenge Score
- Technical Interview*

*Remember that I consider a candidate HIRED when he has both scores greater than or equal to 7; you should apply this logic to get the correct information. How you will handle this data is on you.

And please remember, all the data here is totally random; we used a public library to generate random information.

Data example

candidates.csv									
First Name	Last Name	Email	Application Date	Country	YOE	Seniority	Technology	Code Challenge Score	Technical Interview Score
Bernadette	Langworth	leonard91@yahoo.com	2021-02-26	Norway	2	Intern	Data Engineer	3	3
Camryn	Reynolds	zelda56@hotmail.com	2021-09-09	Panama	10	Intern	Data Engineer	2	10
Larue	Spinka	okey_schultz41@gmail.com	2020-04-14	Belarus	4	Mid-Level	Client Success	10	9
Arch	Spinka	elvera_kulas@yahoo.com	2020-10-01	Eritrea	25	Trainee	QA Manual	7	1
Larue	Altenwerth	minnie.gislason@gmail.com	2020-05-20	Myanmar	13	Mid-Level	Social Media Community Management	9	7
Alec	Abbott	juanita_hansen@gmail.com	2019-08-17	Zimbabwe	8	Junior	Adobe Experience Manager	2	9
Allison	Jacobs	alba_rolfson27@yahoo.com	2018-05-18	Wallis and Futuna	19	Trainee	Sales	2	9
Nya	Skiles	madisen.zulauf@gmail.com	2021-12-09	Myanmar	1	Lead	Mulesoft	2	5
Mose	Lakin	dale_murazik@hotmail.com	2018-03-13	Italy	18	Lead	Social Media Community Management	7	10
Terrance	Zieme	dustin31@hotmail.com	2022-04-08	Timor-Leste	25	Lead	DevOps	2	0
Alyana	Goodwin	vallie.damore@yahoo.com	2019-09-22	Armenia	24	Intern	Development - CMS Backend	4	9

Task breakdown

- Dimensional Data Model (DDM) – Star schema
 - Fact Table.
 - Dimension Tables.
 - Provide the diagram + justification of your design.
- ETL Process
 - Extract: Load the CSV file in Python.
 - Transform:
 - Data cleaning
 - Apply the “HIRED” rule.
 - Load: Insert the transformed data into a DW.
- KPIs & visualizations: Your reports must come from the DW (not the CSV).
 - Hires by technology (pie chart)
 - Hires by year (horizontal bar chart)
 - Hires by seniority (bar chart)
 - Hires by country over years (focus on USA, Brazil, Colombia, and Ecuador) (multiline chart)
- Deliverables in GitHub
 - ETL Notebook (code for Extract, Transform, Load).
 - SQL Queries to extract KPIs from DW.
 - Star Schema Diagram (image + explanation).
 - Visualizations (in notebook or exported as screenshots).
 - README.md explaining your project, setup instructions, and key decisions.
 - .gitignore file.

Evaluation criteria

Item	Description	Weight
Repository & code quality	Repo created, organized, clean structure	0.5

Modeling DDM	Star schema diagram + justification	1.2
DW integration	Correct loading of transformed data into DW; ETL process	1.3
Extracting from DW	Queries pull data from DW, not CSV	0.9
KPIs & visualizations	Correct metrics, accurate, clear charts	0.6
Technical documentation	Approach, setup, usage and explanation of ETL pipeline, challenges, assumptions	0.5